

OptiRisk

Continuous Cyber Risk Quantification (CRQ) & Algorithmic Capital Allocation Platform for Banks, Enterprises, and Critical Infrastructure.

PRIMARY INNOVATION

**FAIR™ Stochastic Loss Modeling
+ Knapsack ROI Optimization**

TARGET SECTOR

**Banking, FinTech, Financial
Institutions & Critical National
Infra**

COMPLIANCE ENGINE

**RBI Cyber Security Framework,
SEBI CSCRf, CERT-In & NIST 2.0**

PLATFORM ARCHITECTURE

**Next.js 14 (App Router) + Firebase
Enterprise RBAC + Monte Carlo
Engine**

Project ID: SIH-OPTIRISK-2024 • Enterprise
Production Standard

Live Platform: <http://localhost:3000> • Repository:
[borse9030/OPTIRISK](https://github.com/borse9030/OPTIRISK)

The Billion-Dollar Disconnect (Problem & Solution)

Why traditional cybersecurity risk assessment fails Indian banks, and how OptiRisk delivers the quantitative standard required by modern regulatory frameworks.

✗ The Legacy Problem in Banks

- ✓ **Vague Heatmaps:** Risk teams present arbitrary "Red, Amber, Green" matrices. A CFO cannot insure or budget against "Red".
- ✓ **CVE Alert Fatigue:** Scanners flag 4,000+ CVEs. CISOs have no mathematical formula to know which flaw exposes the most dollars.
- ✓ **Blind Security Spending:** Budgets are spent based on vendor marketing rather than mathematically proven risk reduction ROI.
- ✓ **Regulatory Penalty Risk:** RBI, SEBI, and CERT-In demand empirical Cyber Risk Quantification and board-level reporting.

✓ The OptiRisk Breakthrough

- ✓ **Dollar-Denominated Loss:** Translates technical CVEs and threat vectors into exact Expected Annual Loss (\$ / ₹) with 95% confidence intervals.
- ✓ **Stochastic FAIR™ Modeling:** Evaluates Threat Event Frequency (TEF), Vulnerability Exploitability, and Primary/Secondary Loss Magnitude.
- ✓ **Algorithmic Portfolio Optimizer:** Employs constrained Knapsack optimization to allocate capital to controls that maximize Risk Reduction ROI.
- ✓ **Automated Regulatory Alignment:** 100% mapped to RBI CSF, SEBI CSCRF, DPDP Act 2023, and NIST CSF 2.0.

🏆 THE 30-SECOND ELEVATOR PITCH FOR SIH INVIGILATORS

"Respected Judges, technical security teams speak in CVEs and CVSS scores, while the Board of Directors and CFOs speak in Rupees, Dollars, and ROI. OptiRisk bridges this gap. We translate cybersecurity vulnerabilities into exact financial loss distributions using the Open FAIR model and 10,000 Monte Carlo iterations. Then, our optimizer algorithm tells the bank exactly where to invest every rupee to achieve maximum risk reduction."

End-to-End System Architecture

LAYER	TECHNOLOGY	ENGINEERING RESPONSIBILITY
Presentation & UI	Next.js 14 App Router, React 18, Tailwind CSS, Lucide Icons, Recharts	High-performance responsive dashboard, real-time loss exceedance curves, 5x5 dynamic risk heatmaps.
Authentication & Gateway	Firebase Auth, Firestore, Strict Next.js Client AuthGuard	Enterprise Role-Based Access Control (CISO, CFO, Risk Lead, SecOps Analyst) with gateway route protection.
Quantitative Engine	Stochastic Monte Carlo Engine (10,000 runs), Open FAIR™ Model	Calculates Threat Event Frequency (TEF), Vulnerability (Vulnerability Likelihood), Loss Magnitude, and VaR (95th percentile).
Capital Optimizer	Pareto Frontier & Bounded Knapsack Optimization Algorithm	Maximizes Delta Risk Reduction subject to CISO budget constraints: $\max \sum \Delta \text{Loss}_i$ subject to $\sum \text{Cost}_i \leq \text{Budget}$.
AI Cyber Copilot	LLM-driven Cyber Analyst with Prompt Engineering & Risk Taxonomy	Instant natural language risk querying, automated Board brief synthesis, and remediation roadmap generation.

Use this exact roadmap during your demonstration. Every single screen includes the navigation route, what to click, and the exact explanation to deliver to the invigilators.

1. Public Landing & Interactive ROI Simulator /

PUBLIC EXPERIENCE

The public enterprise gateway. Features animated metric counters, live cyber threat stats, interactive feature showcase tabs, and a real-time budget simulator widget.

Where to Navigate:

Click the OptiRisk logo or navigate to <http://localhost:3000/>.

What to Click/Interact:

Drag the "Interactive Cyber Budget Allocation Simulator" slider to adjust budget from \$100k to \$1M and watch the risk reduction calculate live!

WHAT TO SAY TO THE JUDGES

"Notice here on our public portal, we immediately demonstrate value with an interactive risk simulator. When an enterprise CFO slides the budget bar, our live algorithm recalculates loss reduction dynamically, showing an average Return on Security Investment (ROSI) of 340%."

2. Enterprise Authentication & Role Assignment Gateway

/login

SECURITY &
RBAC

Enterprise authentication powered by Firebase. Includes dual tabs ("Sign In" and "Create Account") with specialized executive roles: CISO, CFO, Risk Lead, and SecOps Analyst.

Where to Navigate:

Click "Sign In" in the header or go to </login>.

What to Click/Interact:

Click one of the 4 role badges (CISO, CFO, Risk Lead, SecOps). It pre-fills production credentials. Click "Sign In to Platform" to enter.

WHAT TO SAY TO THE JUDGES

"Cyber risk governance requires strict segregation of duties. We implemented Role-Based Access Control: a CFO views financial exposure and capital allocation, while a CISO analyzes threat capabilities and control efficacy. No unauthenticated user can bypass our gateway."

3. Executive Cyber Risk Dashboard

[/dashboard](#)

EXECUTIVE TIER

The C-Suite command center. Displays Total Annualized Loss Expectancy (ALE), Enterprise Risk Score, Return on Security Investment (ROSI), the 5x5 dynamic risk heatmap, and the loss exceedance curve.

Where to Navigate:

Click "Dashboard" in the sidebar or go to [/dashboard](#).

What to Click/Interact:

Point out the 4 KPI cards, hover over the 5x5 Risk Matrix cells to show the count of vulnerabilities, and show the Loss Exceedance Probability chart.

WHAT TO SAY TO THE JUDGES

"This is the core dashboard that modern bank boards need. Instead of qualitative guesses, we present our Annualized Loss Exposure—in this scenario \$2.4 Million—backed by a 95% Value-at-Risk confidence interval. The Loss Exceedance curve models black swan events with mathematically rigorous percentile metrics."

4. Threat Scenario Modeling & TTPs

[/threats](#)

THREAT INTELLIGENCE

Quantifies adversary attack scenarios (Ransomware, SWIFT Fraud, Cloud Account Takeover, Insider Threat) mapped directly to MITRE ATT&CK techniques with Threat Event Frequency (TEF).

Where to Navigate:

Click "Threats" in sidebar or visit [/threats](#).

What to Click/Interact:

Click on threat scenario cards (e.g. *Ransomware Double Extortion*) to inspect Threat Event Frequency (TEF) and Threat Capability (TCap) metrics.

WHAT TO SAY TO THE JUDGES

"Under the FAIR framework, risk is the product of Loss Event Frequency and Loss Magnitude. Here, we model real threat capabilities against our existing defensive controls to derive empirical attack success likelihood."

5. Financial-Impact Vulnerability Prioritization

</vulnerabilities>

VULNERABILITY
OPS

Shifts vulnerability remediation from static CVSS scores to monetary risk ranking. Prioritizes CVEs based on the financial loss exposure of the affected asset.

Where to Navigate:

Click "Vulnerabilities" in sidebar or visit </vulnerabilities>.

What to Click/Interact:

Show CVEs tagged with their Dollar Exposure (e.g., CVE-2023-38606 exposing \$820K) vs technical CVSS score.

WHAT TO SAY TO THE JUDGES

"Standard security teams waste weeks fixing CVSS 9.8 bugs on non-critical test servers while ignoring CVSS 7.5 bugs on core payment APIs. OptiRisk contextualizes CVEs by asset criticality, ensuring engineers patch the flaws that actually protect money first."

6. Critical Asset Value at Risk (VaR)

</assets>

ASSET INVENTORY

Comprehensive inventory of digital and cyber banking assets (Core Banking, SWIFT Connector, Customer KYC Vault, Mobile App API) with assigned financial asset valuation.

Where to Navigate:

Click "Assets" in sidebar or visit </assets>.

What to Click/Interact:

Filter by asset criticality (Tier 1 Mission Critical) and view the replacement value, operational downtime cost per hour, and regulatory data fines.

WHAT TO SAY TO THE JUDGES

"Without asset valuation, cyber risk quantification is impossible. OptiRisk categorizes assets across three loss dimensions: direct asset replacement, operational disruption cost per hour, and statutory DPDP/GDPR privacy penalties."

7. Defensive Controls Catalog & Efficacy Matrix

[/controls](#)

DEFENSIVE
CATALOG

Catalog of preventive and detective security controls (MFA, EDR, Microsegmentation, DLP, Zero Trust) with annual licensing cost and measured risk reduction efficacy.

Where to Navigate:

Click "Controls" in sidebar or visit [/controls](#).

What to Click/Interact:

Review control status (Implemented vs Candidate), cost per user/endpoint, and technical efficacy percentage against ransomware and exfiltration.

WHAT TO SAY TO THE JUDGES

"Every security control has a dollar cost and an efficacy rate. This catalog provides the baseline parameters that feed into our algorithmic investment optimizer."

8. Cyber Investment & Budget Optimizer (FLAGSHIP)

[/optimizer](#)

FLAGSHIP
ALGORITHM

Solves the constrained Knapsack problem to determine the mathematically optimal portfolio of security investments that produces the greatest risk reduction within a given budget.

Where to Navigate:

Click "Optimizer" in sidebar or visit [/optimizer](#).

What to Click/Interact:

Drag the "Available Cybersecurity Budget" slider (e.g. from \$300,000 to \$750,000) and click "Run Optimization". Show how controls are dynamically selected!

WHAT TO SAY TO THE JUDGES

"Judges, this is our core algorithmic innovation! In a real bank, the CISO has a strict budget cap. OptiRisk computes the Pareto-optimal frontier. As I adjust the budget slider, our Knapsack solver determines the exact combination of controls that eliminates the maximum financial exposure per dollar invested."

9. Interactive Monte Carlo Risk Simulation Engine

/simulator

STOCHASTIC
MATH

Executes 10,000 to 50,000 randomized stochastic iterations combining log-normal loss magnitude distributions with Poisson threat arrival rates to generate empirical VaR curves.

Where to Navigate:

Click "Simulator" in sidebar or visit /simulator.

What to Click/Interact:

Select "10,000 Iterations", configure the Confidence Interval (95%), and click "Execute Monte Carlo Simulation". Watch the histogram update in real time!

WHAT TO SAY TO THE JUDGES

"Cyberattacks are low-frequency, high-severity events. Deterministic math fails here. Our Monte Carlo engine simulates 10,000 random enterprise years in milliseconds, plotting the exact probability of exceeding any given loss threshold."

10. Multi-Regulatory Compliance Matrix (RBI / SEBI / CERT-In)

/compliance

INDIAN
REGULATORY

Automated compliance tracker mapping technical controls to RBI Cyber Security Framework (CSF) for Scheduled Commercial Banks, SEBI CSCRF, CERT-In directions, and NIST CSF 2.0.

Where to Navigate:

Click "Compliance" in sidebar or visit /compliance.

What to Click/Interact:

Toggle tabs between "RBI Cyber Security Framework", "SEBI CSCRF", and "NIST CSF 2.0" to show audit readiness scores and remediation gaps.

WHAT TO SAY TO THE JUDGES

"Indian banks face severe regulatory penalties for non-compliance. OptiRisk continuously evaluates technical telemetry against the RBI Master Directions on Cyber Security. What takes audit consulting firms 6 weeks is rendered in OptiRisk in under 2 seconds."

11. Geopolitical Threat Intelligence Radar

</risk-intelligence>

THREAT RADAR

Aggregates global threat intelligence feeds, active APT threat actor profiles (Lazarus, FIN7, LockBit), targeted banking sectors, and live indicators of compromise (IoCs).

Where to Navigate:

Click "Risk Intelligence" in sidebar or visit </risk-intelligence>.

What to Click/Interact:

Review active adversary campaigns targeting financial services and inspect real-time threat feed updates.

WHAT TO SAY TO THE JUDGES

"Our risk calculations are not static. The Risk Intelligence radar ingests global threat telemetry to dynamically calibrate threat frequency parameters when new zero-day campaigns emerge."

12. Global Attack Surface & Threat Vector Map

</map>

GEOSPATIAL OPS

Geospatial visualization of enterprise infrastructure, distributed bank branches, cloud data centers (AWS Mumbai, Azure India), and incoming cyber attack vectors.

Where to Navigate:

Click "Attack Map" in sidebar or visit </map>.

What to Click/Interact:

Inspect distributed nodes and regional threat traffic telemetry originating across international jurisdictions.

WHAT TO SAY TO THE JUDGES

"Provides complete geographic situational awareness. Executive teams can immediately see cross-border cyber threats targeting regional bank endpoints."

13. Autonomous AI Cyber Analyst & Copilot

/analyst

GENAI INNOVATION

Generative AI security analyst assistant. Synthesizes enterprise threat telemetry, explains quantitative loss metrics in boardroom English, and generates executive briefing notes.

Where to Navigate:

Click "AI Analyst" in sidebar or visit /analyst.

What to Click/Interact:

Click one of the suggested prompts (e.g. "Explain our primary ransomware loss drivers" or "Summarize our Q3 RBI compliance gap") and watch the AI respond!

WHAT TO SAY TO THE JUDGES

"Non-technical Board members shouldn't need a PhD in cryptography to understand their exposure. Our AI Copilot allows a CFO to ask plain questions and receive immediate, mathematically grounded strategic recommendations."

14. Executive & Regulatory Audit Reporting

/reports

AUDIT READY

One-click generation of Board of Directors executive briefing decks, RBI supervisory audit documentation, and cyber insurance underwriting filings.

Where to Navigate:

Click "Reports" in sidebar or visit /reports.

What to Click/Interact:

Show the available report packages (Board Pack, RBI Compliance Audit, Cyber Insurance Underwriting Dossier) and export buttons.

WHAT TO SAY TO THE JUDGES

"At the end of the quarter, CISOs spend weeks formatting PowerPoint decks. OptiRisk automates the entire governance output into audit-grade compliance reports with a single click."

15. Governance Settings & Risk Appetite Thresholds

</settings>

GOVERNANCE

Configures organizational risk appetite thresholds, currency localization (INR / USD), alert triggers, and telemetry integration hooks.

Where to Navigate:

Click "Settings" in sidebar or visit </settings>.

What to Click/Interact:

Adjust the "Maximum Acceptable Loss Threshold" and demonstrate how alerts fire when simulations exceed the Board's defined risk tolerance.

WHAT TO SAY TO THE JUDGES

"Every enterprise has a different risk tolerance. Here, executive leadership defines the bank's statutory Loss Appetite, and the platform continuously monitors compliance against this threshold."

The Winning 5–Minute SIH Presentation Script

Practice this exact 5–minute schedule with your team before presenting to the SIH jury.

0:00 -
1:00

Minute 1: The Problem & Executive Gateway

Screen: Public Home (/) and Login (/login)

Script: "Hello Judges. India's banking sector faces unprecedented cyber threats, yet CISOs and CFOs cannot agree on risk because security speaks in CVEs while the board speaks in money. With OptiRisk, we solve this fundamental disconnect. Here is our role-based authentication gateway ensuring strict RBAC for CISO, CFO, and Risk Leads."

1:00 -
2:00

Minute 2: Executive Quantification Dashboard

Screen: Dashboard (/dashboard)

Script: "Once logged in as CISO, we see our quantified cyber exposure: \$2.4M in Expected Annual Loss with a 340% ROSI. Our 5x5 dynamic heatmap and loss exceedance curve show both expected daily losses and 95th-percentile black-swan events using the Open FAIR methodology."

2:00 -
3:00

Minute 3: The Algorithmic Optimizer & Monte Carlo Engine (THE CLIMAX)

Screen: Optimizer (/optimizer) and Simulator (/simulator)

Script: "Now to our flagship innovation: The Cyber Investment Optimizer. When I adjust our available budget slider to \$500k, our Knapsack optimization algorithm automatically selects the optimal bundle of controls to eliminate the maximum risk per dollar. In the simulator, we run 10,000 Monte Carlo iterations in real-time to validate the post-mitigation risk curve."

3:00 -
4:00

Minute 4: RBI / SEBI Regulatory Compliance Matrix

Screen: Compliance (/compliance)

Script: "For banking institutions in India, compliance is mandatory. Our platform maps all defensive controls directly to the RBI Cyber Security Framework, SEBI guidelines, and CERT-In directions, showing instant gap analysis and audit readiness."

4:00 -
5:00

Minute 5: AI Cyber Copilot, Reports & Conclusion

Screen: AI Analyst (/analyst) and Reports (/reports)

Script: "Finally, our AI Copilot allows any executive to query risk exposure in plain English, and our reporting module produces one-click Board-ready dossiers. OptiRisk transforms cybersecurity from a cost center into an optimized, mathematically validated financial investment. Thank you, we are ready for your questions."

JUDGE QUESTION

THE BULLETPROOF WINNING ANSWER

"How do you get loss numbers when cyber breach data is confidential?"

"We use the Open FAIR™ framework with calibrated beta-PERT probability distributions. Rather than guessing a single number, we ask for 90% confidence ranges (minimum, most likely, maximum) calibrated against industry data (Verizon DBIR & Ponemon Institute cost of data breach studies), which the Monte Carlo engine samples across 10,000 iterations."

"How is OptiRisk different from vulnerability scanners like Nessus or Qualys?"

"Nessus and Qualys are discovery tools—they spit out thousands of technical CVEs without business context. OptiRisk sits above them as a quantitative decision engine: we multiply vulnerability exploitability by asset financial value to tell the CFO what financial loss is at stake and which fixes yield the highest ROI."

"Does this align with Indian regulatory frameworks?"

"Yes! We explicitly built automated compliance mapping for the **RBI Cyber Security Framework for Banks**, **SEBI Cybersecurity & Cyber Resilience Framework (CSCRF)**, and India's **DPDP Act 2023** statutory fine schedules."

"What is the mathematical algorithm behind the optimizer?"

"We model capital allocation as a 0/1 Bounded Knapsack Problem solved via Dynamic Programming and greedy Pareto-optimal approximation. The objective function maximizes Total Risk Reduction subject to the budget constraint $\sum c_i x_i \leq B$, where c_i is control cost and $x_i \in \{0, 1\}$."